TASK 6: PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

PROJECT DETAILS

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SCENARIO

The e-commerce bookstore (PALETTE: Pixel & Pages) aims to collect and analyze customer feedback for all purchased titles. Customers submit numerical ratings (1 to 5 stars) along with written review feedback upon completing an order. The Product Review & Rating Management System aggregates customer feedback, evaluates individual title performance, and identifies top-rated titles to drive inventory selection, promotion strategies, and vendor decisions.

1. CREATE REVIEW AND RATING TABLE

i. Review Entity Structure

Attribute	Description	Data Type	Constraints
ReviewID	Unique review identifier	VARCHAR2(10)	Primary Key
CustomerID	Customer reference	VARCHAR2(10)	Foreign Key (customer)
ProductID	Book reference	VARCHAR2(10)	Foreign Key (Book)
Rating	Star rating (1 to 5)	NUMBER(1)	CHECK (Rating BETWEEN 1 AND 5)
ReviewText	Written feedback text	VARCHAR2(100)	-
ReviewDate	Submission date	DATE	-

ii. Review Table Creation Query

```
SQL> CREATE TABLE REVIEW (
  2      ReviewID VARCHAR2(10),
  3      CustomerID VARCHAR2(10),
  4      ProductID VARCHAR2(10),
  5      Rating NUMBER(1),
  6      ReviewText VARCHAR2(100),
  7      ReviewDate DATE,
  8      PRIMARY KEY (ReviewID),
  9      FOREIGN KEY (CustomerID) REFERENCES customer (customer_id),
 10      FOREIGN KEY (ProductID) REFERENCES Book (Book_ID),
 11      CONSTRAINT CHK_Rate CHECK (Rating >= 1 AND Rating <= 5)
 12  );

Table created.
```

```
SQL> DESC Review;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 REVIEWID                                  NOT NULL VARCHAR2(10)
 CUSTOMERID                                         VARCHAR2(10)
 PRODUCTID                                          VARCHAR2(10)
 RATING                                             NUMBER(1)
 REVIEWTEXT                                         VARCHAR2(100)
 REVIEWDATE                                         DATE
```

2. STORE CUSTOMER FEEDBACK AND RATINGS

i. Insert Feedback & Ratings

```
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0001', 'CUST-101', 'B1001', 5, 'Exceptional narrative and world-building!', TO_DATE('2026-01-20', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0002', 'CUST-101', 'B1002', 4, 'Great insights into modern digital art.', TO_DATE('2026-01-21', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0003', 'CUST-102', 'B1004', 5, 'A masterpiece of contemporary fiction.', TO_DATE('2026-02-02', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0004', 'CUST-102', 'B1006', 4, 'Solid explanation of AI concepts.', TO_DATE('2026-02-03', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0005', 'CUST-103', 'B1005', 5, 'Hands-down the best design textbook.', TO_DATE('2026-02-15', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0006', 'CUST-104', 'B1008', 4, 'Very helpful code examples included.', TO_DATE('2026-02-28', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0007', 'CUST-104', 'B1006', 5, 'Clear layout and brilliant illustrations.', TO_DATE('2026-03-01', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0008', 'CUST-105', 'B1010', 4, 'Thorough breakdown of historical events.', TO_DATE('2026-03-12', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0009', 'CUST-106', 'B1009', 5, 'Crisp writing and fast-paced story.', TO_DATE('2026-03-25', 'YYYY-MM-DD'));

1 row created.
```

```
SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0010', 'CUST-105', 'B1015', 4, 'Informative, but binding could be better.', TO_DATE('2026-03-26', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0011', 'CUST-107', 'B1008', 5, 'Essential reading for CS students!', TO_DATE('2026-04-10', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0012', 'CUST-108', 'B1012', 4, 'Reliable reference guide for SQL.', TO_DATE('2026-04-30', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0013', 'CUST-108', 'B1006', 2, 'Average quality, missing index pages.', TO_DATE('2026-05-01', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0014', 'CUST-110', 'B1012', 5, 'Extremely comprehensive and useful.', TO_DATE('2026-05-28', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0015', 'CUST-111', 'B1009', 4, 'Great sequel, lived up to expectations.', TO_DATE('2026-06-15', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0016', 'CUST-111', 'B1015', 5, 'Top-notch print and content quality.', TO_DATE('2026-06-16', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0017', 'CUST-112', 'B1014', 4, 'Practical exercises are well structured.', TO_DATE('2026-07-01', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Review (ReviewID, CustomerID, ProductID, Rating, ReviewText, ReviewDate)
  2  VALUES ('REV0018', 'CUST-112', 'B1003', 5, 'Stunning visuals and typography.', TO_DATE('2026-07-02', 'YYYY-MM-DD'));

1 row created.
```

ii. Display Review Table Contents

```
SQL> SELECT * FROM Review;

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
REV0001    CUST-101   B1001               5
Exceptional narrative and world-building!
20-JAN-26

REV0002    CUST-101   B1002               4
Great insights into modern digital art.
21-JAN-26

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------

REV0003    CUST-102   B1004               5
A masterpiece of contemporary fiction.
02-FEB-26

REV0004    CUST-102   B1006               4
Solid explanation of AI concepts.

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
03-FEB-26

REV0005    CUST-103   B1005               5
Hands-down the best design textbook.
15-FEB-26

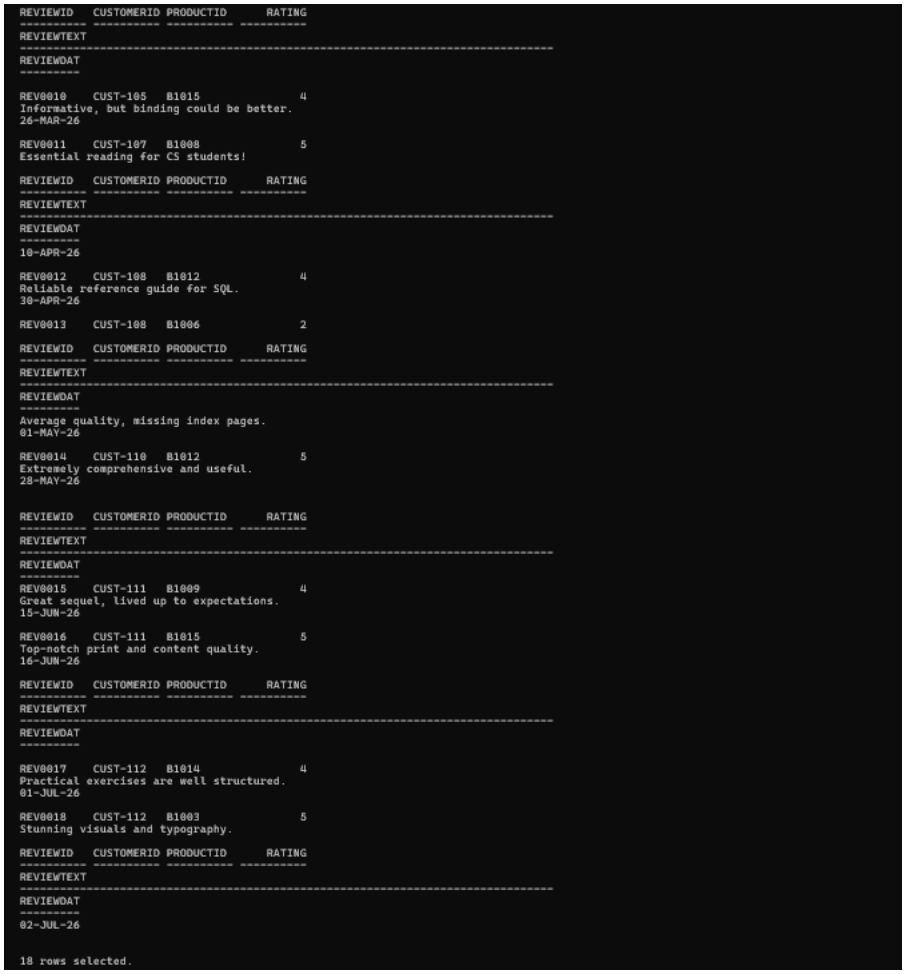
REV0006    CUST-104   B1008               4

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
Very helpful code examples included.
28-FEB-26

REV0007    CUST-104   B1006               5
Clear layout and brilliant illustrations.
01-MAR-26


REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
REV0008    CUST-105   B1010               4
Thorough breakdown of historical events.
12-MAR-26

REV0009    CUST-106   B1009               5
Crisp writing and fast-paced story.
25-MAR-26
```



iii. Enforce Rating Constraints (1 to 5 Stars)

```
SQL> ALTER TABLE Review
  2  ADD CONSTRAINT CHK_RATE
  3  CHECK (Rating >= 1 AND Rating <= 5);

Table altered.
```

```
SQL> INSERT INTO Review
  2  VALUES ('REV0019',
  3          'CUST-101',
  4          'B1001',
  5          6,
  6          'Over the top excellence!',
  7          SYSDATE);
INSERT INTO Review
*
ERROR at line 1:
ORA-02290: check constraint (C##KAVIYA25.CHK_RATE) violated
```

iv. Update Review Comments

```
SQL> UPDATE Review
  2       SET ReviewText = 'Good quality overall, but printing could be improved'
  3       WHERE ReviewID = 'REV0013';

1 row updated.
```

v. Remove Inappropriate/Invalid Reviews

```
SQL> DELETE FROM Review
  2       WHERE ReviewID = 'REV0018';

1 row deleted.
```

3. RETRIEVE PRODUCT REVIEW DETAILS

i. Display all reviews for a specific product (B1012)

```
SQL> SELECT * FROM Review WHERE ProductID = 'B1012';

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
REV0012    CUST-108   B1012               4
Reliable reference guide for SQL.
30-APR-26

REV0014    CUST-110   B1012               5
Extremely comprehensive and useful.
28-MAY-26

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
```

ii. Display customer name with their corresponding reviews

```
SQL> SELECT
  2      c.first_name || ' ' || c.last_name AS Customer_Name,
  3      r.ReviewID,
  4      r.ProductID,
  5      r.Rating,
  6      r.ReviewText
  7  FROM customer c
  8  JOIN Review r ON c.customer_id = r.CustomerID;

CUSTOMER_NAME                          REVIEWID   PRODUCTID      RATING
-------------------------------------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
Ananya Roy                             REV0001    B1001               5
Exceptional narrative and world-building!

Ananya Roy                             REV0002    B1002               4
Great insights into modern digital art.

Rohan Mehta                            REV0003    B1004               5
A masterpiece of contemporary fiction.


CUSTOMER_NAME                          REVIEWID   PRODUCTID      RATING
-------------------------------------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
Rohan Mehta                            REV0004    B1006               4
Solid explanation of AI concepts.

Priya Nair                             REV0005    B1005               5
Hands-down the best design textbook.

Vikram Seth                            REV0006    B1008               4
Very helpful code examples included.


CUSTOMER_NAME                          REVIEWID   PRODUCTID      RATING
-------------------------------------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
Vikram Seth                            REV0007    B1006               5
Clear layout and brilliant illustrations.

Aarav Sharma                           REV0008    B1010               4
Thorough breakdown of historical events.

Ishita Menon                           REV0009    B1009               5
Crisp writing and fast-paced story.
```

```
CUSTOMER_NAME                          REVIEWID   PRODUCTID      RATING
-------------------------------------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
Aarav Sharma                           REV0010    B1015               4
Informative, but binding could be better.

Kabir Singh                            REV0011    B1008               5
Essential reading for CS students!

Diya Patel                             REV0012    B1012               4
Reliable reference guide for SQL.


CUSTOMER_NAME                          REVIEWID   PRODUCTID      RATING
-------------------------------------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
Diya Patel                             REV0013    B1006               2
Good quality overall, but printing could be improved

Meera Rao                              REV0014    B1012               5
Extremely comprehensive and useful.

Rohan Verma                            REV0015    B1009               4
Great sequel, lived up to expectations.


CUSTOMER_NAME                          REVIEWID   PRODUCTID      RATING
-------------------------------------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
Rohan Verma                            REV0016    B1015               5
Top-notch print and content quality.

Anika Iyer                             REV0017    B1014               4
Practical exercises are well structured.


17 rows selected.
```

iii. Find products having maximum reviews

```
SQL> SELECT
  2      b.Book_ID,
  3      b.Title AS Book_Title,
  4      COUNT(r.ReviewID) AS Review_Count
  5  FROM Book b
  6  JOIN Review r ON b.Book_ID = r.ProductID
  7  GROUP BY b.Book_ID, b.Title
  8  ORDER BY Review_Count DESC;

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
B1006
Gone Girl
           3

B1012
The Hobbit
           2

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------

B1008
Murder on the Orient Express
           2

B1009
Pride and Prejudice

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
           2

B1015
Wings of Fire
           2

B1010
```

```
BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
Me Before You
           1

B1002
Dune
           1


BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
B1014
The Shining
           1

B1005
The Silent Patient
           1

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------

B1004
Designing Data-Intensive Applications
           1

B1001
The Great Gatsby

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
           1


11 rows selected.
```

iv. Display recent customer feedback

```
SQL>
SQL> SELECT *
  2  FROM
  3  (
  4      SELECT
  5          c.first_name || ' ' || c.last_name AS Customer_Name,
  6          b.Title AS Book_Title,
  7          r.Rating,
  8          r.ReviewText AS Customer_Feedback,
  9          r.ReviewDate AS Review_Date
 10      FROM Review r
 11      JOIN Customer c
 12          ON r.CustomerID = c.Customer_ID
 13      JOIN Book b
 14          ON r.ProductID = b.Book_ID
 15      ORDER BY r.ReviewDate DESC
 16  )
 17  WHERE ROWNUM <= 5;

CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Anika Iyer
The Shining
         4

CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Practical exercises are well structured.
01-JUL-26
```

```
CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Rohan Verma
Wings of Fire
         5

CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Top-notch print and content quality.
16-JUN-26


CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Rohan Verma
Pride and Prejudice
         4
```

```
CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Great sequel, lived up to expectations.
15-JUN-26


CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Meera Rao
The Hobbit
         5

CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Extremely comprehensive and useful.
28-MAY-26
```

```
CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Diya Patel
Gone Girl
         2

CUSTOMER_NAME
----------------------------------------
BOOK_TITLE
--------------------------------------------------------------------------------
    RATING
----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
Good quality overall, but printing could be improved
01-MAY-26
```

v. Retrieve reviews with ratings above 4

```
SQL> SELECT * FROM Review WHERE Rating > 4;

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
REV0001    CUST-101   B1001               5
Exceptional narrative and world-building!
20-JAN-26

REV0003    CUST-102   B1004               5
A masterpiece of contemporary fiction.
02-FEB-26

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------

REV0005    CUST-103   B1005               5
Hands-down the best design textbook.
15-FEB-26

REV0007    CUST-104   B1006               5
Clear layout and brilliant illustrations.

REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
01-MAR-26

REV0009    CUST-106   B1009               5
Crisp writing and fast-paced story.
25-MAR-26

REV0011    CUST-107   B1008               5
```

```
REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
Essential reading for CS students!
10-APR-26

REV0014    CUST-110   B1012               5
Extremely comprehensive and useful.
28-MAY-26


REVIEWID   CUSTOMERID PRODUCTID      RATING
---------- ---------- ---------- ----------
REVIEWTEXT
--------------------------------------------------------------------------------
REVIEWDAT
---------
REV0016    CUST-111   B1015               5
Top-notch print and content quality.
16-JUN-26


8 rows selected.
```

4. CALCULATE AVERAGE PRODUCT RATINGS

i. Calculate average rating for each product

```
SQL> SELECT
  2      b.Title AS Book_Title,
  3      ROUND(AVG(r.Rating), 2) AS Average_Rating
  4  FROM Book b
  5  JOIN Review r ON b.Book_ID = r.ProductID
  6  GROUP BY b.Title;

BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
Gone Girl
          3.67

Dune
             4

The Silent Patient
             5


BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
Pride and Prejudice
           4.5

Murder on the Orient Express
           4.5

Designing Data-Intensive Applications
             5


BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
The Great Gatsby
             5

Me Before You
             4

Wings of Fire
           4.5
```

```
BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
The Hobbit
           4.5

The Shining
             4


11 rows selected.
```

ii. Count the number of reviews per product

```
SQL> SELECT
  2      b.Title AS Book_Title,
  3      COUNT(r.Rating) AS Count_Rating
  4  FROM Book b
  5  JOIN Review r ON b.Book_ID = r.ProductID
  6  GROUP BY b.Title;

BOOK_TITLE
--------------------------------------------------------------------------------
COUNT_RATING
------------
Gone Girl
           3

Dune
           1

The Silent Patient
           1


BOOK_TITLE
--------------------------------------------------------------------------------
COUNT_RATING
------------
Pride and Prejudice
           2

Murder on the Orient Express
           2

Designing Data-Intensive Applications
           1


BOOK_TITLE
--------------------------------------------------------------------------------
COUNT_RATING
------------
The Great Gatsby
           1

Me Before You
           1

Wings of Fire
           2
```

```
BOOK_TITLE
--------------------------------------------------------------------------------
COUNT_RATING
------------
The Hobbit
           2

The Shining
           1


11 rows selected.
```

iii. Find highest-rated products

```
SQL>
SQL> SELECT
  2      b.Book_ID,
  3      b.Title AS Book_Title,
  4      ROUND(AVG(r.Rating), 2) AS Average_Rating
  5  FROM Book b
  6  JOIN Review r ON b.Book_ID = r.ProductID
  7  GROUP BY b.Book_ID, b.Title
  8  HAVING AVG(r.Rating) = (
  9      SELECT MAX(Avg_Rating)
 10      FROM (
 11          SELECT AVG(Rating) AS Avg_Rating
 12          FROM Review
 13          GROUP BY ProductID
 14      )
 15  );

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
B1004
Designing Data-Intensive Applications
             5

B1005
The Silent Patient
             5

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------

B1001
The Great Gatsby
             5
```

iv. Identify products with average rating above 4

```
SQL> SELECT
  2      b.Book_ID,
  3      b.Title AS Book_Title,
  4      ROUND(AVG(r.Rating), 2) AS Average_Rating
  5  FROM Book b
  6  JOIN Review r ON b.Book_ID = r.ProductID
  7  GROUP BY b.Book_ID, b.Title
  8  HAVING AVG(r.Rating) > 4
  9  ORDER BY Average_Rating DESC;

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
B1001
The Great Gatsby
             5

B1005
The Silent Patient
             5

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------

B1004
Designing Data-Intensive Applications
             5

B1012
The Hobbit

BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
           4.5

B1008
Murder on the Orient Express
           4.5

B1009
```

```
BOOK_ID
-------
BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
Pride and Prejudice
           4.5

B1015
Wings of Fire
           4.5


7 rows selected.
```

5. IDENTIFY HIGHLY RATED PRODUCTS (BUSINESS REPORTS)

Report 1: Product Rating Analysis

- **Product name**
- **Number of reviews**
- **Average rating**

```
SQL> SELECT
  2      b.Title AS Book_Title,
  3      COUNT(r.ReviewID) AS Number_of_Reviews,
  4      ROUND(AVG(r.Rating), 2) AS Average_Rating
  5  FROM Book b
  6  LEFT JOIN Review r ON b.Book_ID = r.ProductID
  7  GROUP BY b.Book_ID, b.Title
  8  ORDER BY Average_Rating DESC NULLS LAST;

BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
The Silent Patient
                1              5

Designing Data-Intensive Applications
                1              5

The Great Gatsby
                1              5


BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
The Hobbit
                2            4.5

Murder on the Orient Express
                2            4.5

Pride and Prejudice
                2            4.5


BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Wings of Fire
                2            4.5

Dune
                1              4

The Shining
                1              4
```

```
BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Me Before You
                1              4

Gone Girl
                3           3.67

The Diary of a Young Girl
                0


BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
The Da Vinci Code
                0

Harry Potter and the Philosopher's Stone
                0

Sapiens
                0


BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Steve Jobs
                0

Treasure Island
                0

Around the World in 80 Days
                0
```

```
BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Sherlock Holmes
                0

The Fault in Our Stars
                0

Percy Jackson and the Lightning Thief
                0


BOOK_TITLE
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Angels and Demons
                0

Foundation
                0

Dracula
                0


24 rows selected.
```

Report 2: Customer Feedback Analysis

a. Most reviewed products

```
SQL> SELECT
  2      b.Title AS Book_Title,
  3      COUNT(r.ReviewID) AS Review_Count
  4  FROM Book b
  5  JOIN Review r ON b.Book_ID = r.ProductID
  6  GROUP BY b.Book_ID, b.Title
  7  ORDER BY Review_Count DESC;

BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
Gone Girl
           3

The Hobbit
           2

Murder on the Orient Express
           2


BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
Pride and Prejudice
           2

Wings of Fire
           2

Me Before You
           1


BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
Dune
           1

The Shining
           1

The Silent Patient
           1
```

```
BOOK_TITLE
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
Designing Data-Intensive Applications
           1

The Great Gatsby
           1


11 rows selected.
```

b. Highly rated products

```
SQL> SELECT
  2      b.Title AS Book_Title,
  3      ROUND(AVG(r.Rating), 2) AS Average_Rating
  4  FROM Book b
  5  JOIN Review r ON b.Book_ID = r.ProductID
  6  GROUP BY b.Book_ID, b.Title
  7  HAVING AVG(r.Rating) > 4
  8  ORDER BY Average_Rating DESC;

BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
The Great Gatsby
             5

The Silent Patient
             5

Designing Data-Intensive Applications
             5


BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
The Hobbit
           4.5

Murder on the Orient Express
           4.5

Pride and Prejudice
           4.5


BOOK_TITLE
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
Wings of Fire
           4.5


7 rows selected.
```

c. Products requiring improvement

```
SQL> SELECT
  2      b.Title AS Book_Title,
  3      COUNT(r.ReviewID) AS Review_Count,
  4      ROUND(AVG(r.Rating), 2) AS Average_Rating
  5  FROM Book b
  6  JOIN Review r ON b.Book_ID = r.ProductID
  7  GROUP BY b.Book_ID, b.Title
  8  HAVING AVG(r.Rating) < 3
  9  ORDER BY Average_Rating;

no rows selected
```

Report 3: Rating Distribution Analysis

a. Complete Rating Breakdown

- **Number of 5-Star Ratings**

```
SQL> SELECT COUNT(*) AS FIVE_STAR_RATINGS
  2  FROM REVIEW
  3  WHERE RATING = 5;

FIVE_STAR_RATINGS
-----------------
                8
```

- **Number of 4-Star Ratings**

```
SQL> SELECT COUNT(*) AS FOUR_STAR_RATINGS
  2  FROM REVIEW
  3  WHERE RATING = 4;

FOUR_STAR_RATINGS
-----------------
                8
```

- **Number of Low-Rated Products**

```
SQL> SELECT COUNT(*) AS LOW_RATED_PRODUCTS
  2  FROM
  3  (
  4      SELECT PRODUCTID
  5      FROM REVIEW
  6      GROUP BY PRODUCTID
  7      HAVING AVG(RATING) < 3
  8  );

LOW_RATED_PRODUCTS
------------------
                 0
```

- **Complete Rating Distribution**

```
SQL> SELECT
  2      RATING,
  3      COUNT(*) AS NUMBER_OF_RATINGS
  4  FROM REVIEW
  5  GROUP BY RATING
  6  ORDER BY RATING DESC;

    RATING NUMBER_OF_RATINGS
---------- -----------------
         5                 8
         4                 8
         2                 1
```

b. Summary Metrics

- **Number of 5-Star Ratings: 8**
- **Number of 4-Star Ratings: 8**
- **Low-Rated Titles (Avg Rating $\leq$ 3): 0**